Notes on grifting by optimal transport

Gergely Neu GERGELY.NEU@GMAIL.COM
Universitat Pompeu Fabra, Barcelona, Spain

Lorenzo Croissant LORENZO.CROISSANT@UPF.EDU
Universitat Pompeu Fabra, Barcelona, Spain

Keywords: optimal transport, generative modeling, Markov chains

Consider an affine measurable space $\mathcal{S} \subset \mathbb{R}^d$. Fixing $(\Omega, \Sigma, \mathbb{P})$, $\Omega = \mathcal{S}$, $\Sigma = \mathscr{B}(\mathcal{S})$ the Borell σ-algebra of $\mathcal{S}$, and $\mathbb{P}$ a probability measure on (Ω, Σ). Let $v := (v_t)_{t \in \mathbb{N}}$ be a collection of Σ-measurable automorphisms on $\mathcal{S}$.

The dynamical system of a drifting model (Deng et al., 2026) evolves the prediction during training according to the following Markov chain:

$$X_{t+1}^v = X_t^v + v_t(X_t) \in \mathcal{S} \tag{1}$$

equivalently, letting μ_0 be probability measure on (Ω, Σ), this can be viewed in terms of the evolution of the law of X_t from its initial distribution μ_0 as

$$\begin{cases} \mathcal{L}(X_{t+1}^v) = (\mathrm{Id} + v_t)|_\sharp \mathcal{L}(X_t^v) \\ \mathcal{L}(X_0^v) = \mu_0 \end{cases} \tag{2}$$

The development of drifting models Deng et al. (2026) rests on the idea of crafting (by hand) a vector-field-valued map v which is asymptotically stable and whose attracting equilibrium induces $X_\infty \sim \nu$ for a given target distribution $\nu \in \mathscr{P}(\mathcal{S})$. By using the methodology of least-squares fitting outlined in Deng et al. (2026), the specification of v can be leveraged into the *training* of a neural network f which yields an approximate pushforward map transforming a given noise distribution μ_0 into the target distribution ν, i.e. $\nu \approx f|_\sharp \mu_0$.

From a technical standpoint, the authors of Deng et al. (2026) construct a family of vector fields v depending on ν and $\mathcal{L}(X_t)$ which admits as *an* equilibrium the target distribution ν. However, under general conditions it remains unclear whether the resulting mean-field dynamics of X_t will meaningfully converge to ν as $t \to \infty$, despite them giving some sufficient conditions.

A contrario, it is tempting to reinterpret (1) as a control problem, and try and determine v through the specification of an objective function to be optimised. At an intuitive level, we can then view the efforts of Deng et al. (2026) as an ad-hoc attempt to guess a control v which will yield the desired target distribution ν.

Inspired by the recent work of Müller et al. (2026), we propose to study this as a *measure constrained* optimal control problem.

1. Proposed setting

Let $\gamma \in [0, 1)$, $c : \mathcal{S}^2 \to \mathbb{R}$ be a measurable cost function, and let $\mu_0 \in \mathscr{P}(\mathcal{S})$ be a given initial distribution, and consider the optimal control problem,

$$\begin{aligned} J_p := \min_{v \in \mathcal{V}} \quad & \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t c(X_t^v, X_{t+1}^v)\right] \\ \text{subject to } & X_{t+1}^v = X_t^v + v_t(X_t) \text{ for all } t \geq 0, \\ & X_0 \sim \mu_0, \end{aligned} \tag{3}$$

where $\mathcal{V} \subset (\mathcal{S}^{\mathcal{S}})^{\mathbb{N}}$ is a set of admissible controls. This problem is highly non-linear and non-smooth in v in general, which complicates resolution. While the primal theory which tackles (3) head-on, we will instead focus on the (control-theoretic) dual formulation of the problem, which is linear in the space of measures.

To do so, let $\mathscr{M}_+(\mathcal{S}^2)$ be the space of σ-finite positive measures on $\mathcal{S}^2$, and let $\mathscr{P}(\mathcal{S}^2) \subset \mathscr{M}(\mathcal{S}^2)$ be the space of probability measures on $\mathcal{S}^2$. We define the "forward" and "backward" marginalisation operators $\mathcal{F}$ and $\mathcal{B}$ on $\mathscr{M}_+(\mathcal{S}^2)$ as follows:

$$\mathcal{F} : \mu \in \mathscr{M}_+(\mathcal{S}^2) \to \int_{\mathcal{S}} \mu(\mathrm{d}x, \cdot) \in \mathscr{M}_+(\mathcal{S}) \tag{4}$$

$$\mathcal{B} : \mu \in \mathscr{M}_+(\mathcal{S}^2) \to \int_{\mathcal{S}} \mu(\cdot, \mathrm{d}y) \in \mathscr{M}_+(\mathcal{S}) \tag{5}$$

For any $\gamma \in [0, 1)$, let $\mathscr{P}_\gamma(\mathcal{S}^2) := (1-\gamma)^{-1}\mathscr{P}(\mathcal{S}^2)$ be the subspace of $\mathscr{M}_+(\mathcal{S}^2)$ of measures with total mass $(1-\gamma)^{-1}$.

For any $v \in \mathcal{V}$, with μ_0 implicitely fixed, let $\mu^v \in \mathscr{P}_\gamma(\mathcal{S}^2)$ be the measure defined by

$$\mu^v(\mathrm{d}x, \mathrm{d}y) := \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_t^v, X_{t+1}^v)(\mathrm{d}x, \mathrm{d}y). \tag{6}$$

This is equivalently written as

$$\mu^v : (A, B) \in \Sigma^2 \to \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(X_t^v \in A, X_{t+1}^v \in B) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(X_t^v \in A, (\mathrm{Id} + v_t)(X_t^v) \in B)\,. \tag{7}$$

The *control-theoretic dual* is the linear problem

$$J_d := \min_{v \in \mathcal{V}} \quad \int c(x, y) \mu^v(\mathrm{d}x, \mathrm{d}y) \tag{8}$$

subject to $\mu^v \in \mathscr{P}_\gamma(\mathcal{S}^2)$,

$$\mathcal{B}\mu^v = \mu_0 + \gamma \mathcal{F}\mu^v \qquad\qquad \mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \bigcirc_{s=1}^{t} (\mathrm{Id} + v_s)|_\sharp \mu_0$$

Lemma 1 *The control-theoretic duality is strong $J_p = J_d$. (We will treat minimiser equivalence later.)*

Proof That $\int c \mathrm{d}\mu^v = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t c(X_t^v, X_{t+1}^v)]$ is immediate from the definition of μ^v.

To establish the constraints of the dual, note that, by definition $\mathcal{B}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_t^v)$, and thus

$$\mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_{t+1}^v) = \frac{1}{\gamma}(\mathcal{B}\mu^v - \mu_0). \tag{9}$$

This establishes the constraint associated to $X_0 \sim \mu_0$. The constraint for the dynamics directly leads to

$$\mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_{t+1}^v) = \sum_{t=0}^{\infty} \gamma^t (\mathrm{Id} + v_t)|_\sharp \mathcal{L}(X_t^v) = \sum_{t=0}^{\infty} \gamma^t \bigcirc_{s=1}^{t} (\mathrm{Id} + v_s)|_\sharp \mu_0\,. \tag{10}$$

■

References

M. Deng, H. Li, T. Li, Y. Du, and K. He. Generative modeling via drifting. *arXiv preprint arXiv:2602.04770*, 2026.

A. Müller, P. Moreno, and G. Neu. Unpublished work, 2026.